Smith's Propositions 2 and 4, natively: run report

Run `run-20261014T1827Z` (UTC start) · study `research/smith_prop24_native_20261014` · descriptive, no verdicts

What ran. `examples/smith_prop24.rs` built with `cargo build --release --example smith_prop24` at commit `31d0beec8` (the protocol freeze commit; `git status` of `src/` and `examples/` recorded as clean at the end of all 4 runs). `PROTOCOL.md` SHA-256 `d7cbe486ff79fe60`..., the same in every manifest entry. Host: 4-vCPU Intel Xeon @ 2.80 GHz cloud container, 15 GB RAM, Linux 6.18, `rustc 1.97.0 (2d8144b78 2026-07-07)`, rayon with 4 threads, shared with another agent's builds, no isolation. Wall times are incidental; the results are exact computations and counts, not timings.

run	command (after the binary)	seconds	status
control_example1	`control --out research/smith_prop24_native_20261014/data`	0	completed
random_p2147483647	`random --p 2147483647 --n 1000000 --seed 20261014 --out research/smith_prop24_native_20261014/data`	81.7	completed
census_p7	`census --p 7 --perms 199 --seed 20261014 --out research/smith_prop24_native_20261014/data`	15.6	completed
census_p11	`census --p 11 --perms 199 --seed 20261014 --out research/smith_prop24_native_20261014/data`	892.1	completed

Arms and seeds. Control: Smith's Example 1 over $\mathbb{F}_{37}$ and the same curve in a degree-8 model. Question 1: $N = 1000000$ random squarefree degree-8 models over $p = 2^{31} - 1$, seed 20261014 (`SplitMix64`, curve i from `seed XOR i·0xD1B54A32D192ED03`). Question 2: every monic octic with zero x^7 coefficient over $\mathbb{F}_7$ (823543 codes) and $\mathbb{F}_{11}$ (19487171 codes), each squarefree one as a curve and its twist by 3 (p = 7) and 2 (p = 11); 199 permutations of the independence null with seed 20261014. Requested runs: 4. Completed: 4. Failed or timed out: none. After the runs, `smith_prop24 resummarize` rebuilt every summary byte for byte from the rows, and `verdict` and `describe` (post hoc, protocol amendment 1) read only the recorded files.

Measured values (unit in each row; intervals are those the summary recorded).

Q1 fraction	k / n	value	98.75% Clopper–Pearson	prediction	Smith Ex. 2
curves with a rational tractable subgroup	501874 / 1000000	0.50187	[0.50062, 0.50312]	0.50159	0.50201
tractable subgroups with a rational trigonal map (Prop. 2)	500560 / 1001128	0.5	[0.49875, 0.50124]	0.5	0.50051
of those, subgroups with a rational isogeny (Prop. 4)	250793 / 500560	0.50102	[0.49926, 0.50279]	0.5	0.49949
curves with full_success	186382 / 1000000	0.18638	[0.18541, 0.18736]	0.18574	0.18581

p	curves	classes	direct full_success D	class-level C	ceiling U	r	null C (2.5–97.5%)	tractable / full classes
7	1411788	5231	0.1758	0.4304	0.4897	0.811	0.4897–0.4897	1855 / 1295
11	35431220	21701	0.1805	0.475	0.4943	0.938	0.4943–0.4943	7431 / 6221

Instrument diagnostics (counts over all subgroups or maps of the run; the protocol requires the fail, mismatch, disagreement and error columns to be 0).

run	maps	rank M < 4	disc = 0	not normal form	deg g < 3	construction failed	pair fail	eq. (7) vs norm	Lemma 4 fail	λ disagree	X agree / disagree / n.a. / error
Q1	1001120	0	0	0	0	0	0	0	0	0	0 / 0 / 0 / 0
p = 7	608736	12570	77346	79560	55290	27645	0	0	0	0	536040 / 0 / 570852 / 0
p = 11	16166860	134110	1357050	1374800	1070870	535435	0	0	0	0	17845720 / 0 / 12346260 / 0

Controls. Example 1 passed: yes (one rational tractable subgroup, Smith's N, D among the computed maps, $\delta_0, \delta_2, \delta_4, 64s = \delta_1^2, f_0, f_1, f_2$ and $\#H(\mathbb{F}_{37}) = 42$ all as printed by Smith, Prop. 4 rational in both models). Frozen 2026-10-13 counts reproduced: p = 7 yes, p = 11 yes. Class closure of "has a rational tractable subgroup" equals its direct fraction: p = 7 yes, p = 11 yes.

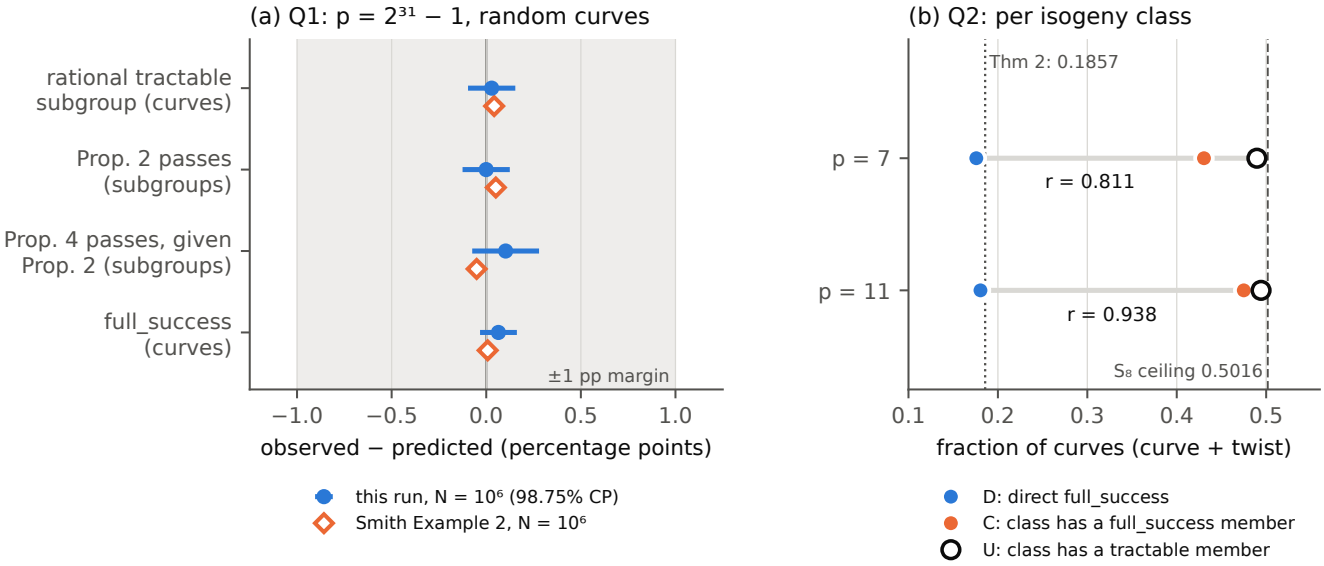


Figure 1: Recorded values of this run. (a) Question 1, observed minus predicted, 98.75% Clopper–Pearson intervals. (b) Question 2, exhaustive: D, C, U per prime. Source: `figures/plot.py` from `data/*.json`.

Raw records. `data/random_p2147483647_rows.csv.gz` (one row per curve), `data/census_p{7,11}_classes.csv` (one row per class), `data/census_p{7,11}_types.csv`, `data/census_p{7,11}_raw.json`, the summaries `data/*_summary.json`, `data/control_example1.json`, `data/run_manifest.json`. Interpretation and verdicts are in `REPORT.pdf`, not here.